NUMERICAL ANALYSIS FOR MACHINE LEARNING PROJECT

TRUSTWORTHY PNEUMONIA DETECTION IN CHEST X-RAY IMAGING THROUGH ATTENTION-GUIDED DEEP LEARNING

Emanuele De Simone

SCOPE OF THE PROJECT

- To replicate the model described in the scientific paper “Trustworthy pneumonia detection in chest X-ray imaging through attention-guided deep learning” by Houmem Slimi, Ala Balti, Sabeur Abid and Mounir Sayadi on Nature

- To use spiking neural networks (SNN), trying to find a way to somehow transform static X-ray images into dynamic time-dependant data

- To report the main experimental results to see whether the trained model achieves sufficient enough performances

MATCHEMATICAL MODELS

- **SNN:**

Membrane Potential Update: $$V\left(t+1\right)=\beta V\left(t\right)+\left(1-\beta\right)I\left(t\right)$$

Spike Generation: $$Spike\left(t\right)=\begin{cases}1 & \text{if } V(t)\geq V_{\text{thresh}}\\ 0 & \text{otherwise}\end{cases}$$

Reset Mechanism: $$V(t)=V_{\text{reset}}\text{ if Spike}(t)=1$$

Output: $$\text{Output}=\frac{1}{T}\sum_{t=1}^{T}\text{Spike}(t)$$

MATHEMATICAL MODELS

- **Bidirectional Gated Recurrent Unit (BiGRU):**

Update Gate: $$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$$

Reset Gate: $$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$

Candidate Hidden State: $$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h)$$

Hidden State: $$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

MATHEMATICAL MODELS

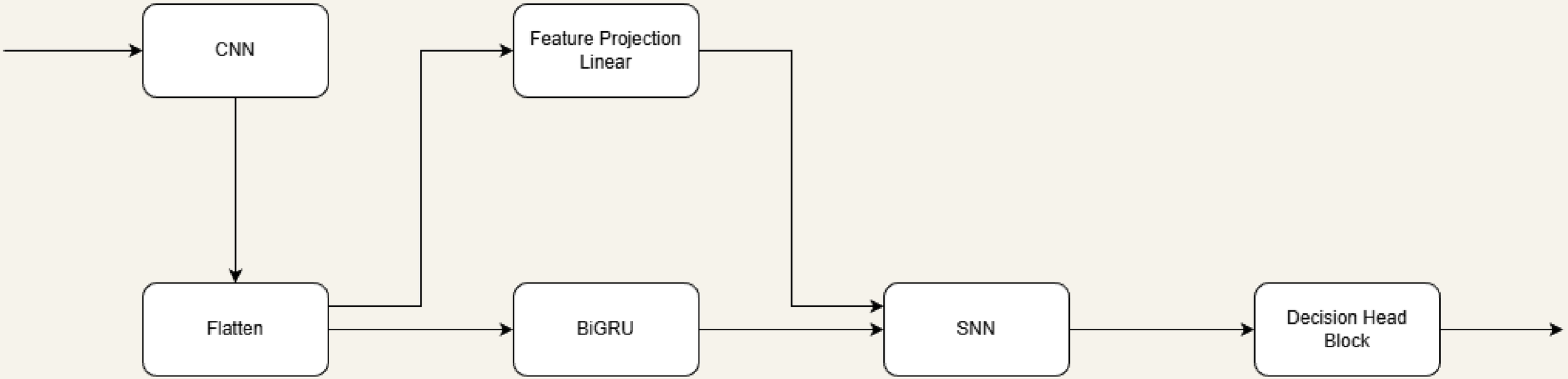
- **BiGRU:**

  it transforms the static spatial features of the image into a temporal sequence

- **SNN:**

  it uses the output of the BiGRU to classify the images into the two classes

NEURAL NETWORK ARCHITECTURE



TECHNOLOGIES USED

- **Python** as the programming language
- **kaggle.com** to run the notebook with the **GPU P100** as hardware accelerator
- **PyTorch** as the framework to train the neural network
- **numpy, matplotlib, sklearn...**

EXPERIMENTAL SETUP

- **PREPROCESSING** to transorm images innto matrices
- **CROSS-VALIDATION (5-fold)** on learning rate and decay factor of the SNN for model selection
- **TRAINING**
- **TESTING**

CROSS-VALIDATION AND TRAINING

- **Hyperparameters:** epochs = 30, batch_size = 32, weight_decay = 1e-5

- A**ugmentation** (Only in training)**:** each time an image is extracted from the training set, there is a 70% probability that a random transformation is performed

- O**ptimization:** the optimization algorithm chosen is Adam with a OneCycleLR scheduler and a cross entropy loss function with a penalization on the loss for the pneumonia class to address the fact that the dataset is imbalanced.

TESTING

- **Test Set:** the model has been tested on new unseen data

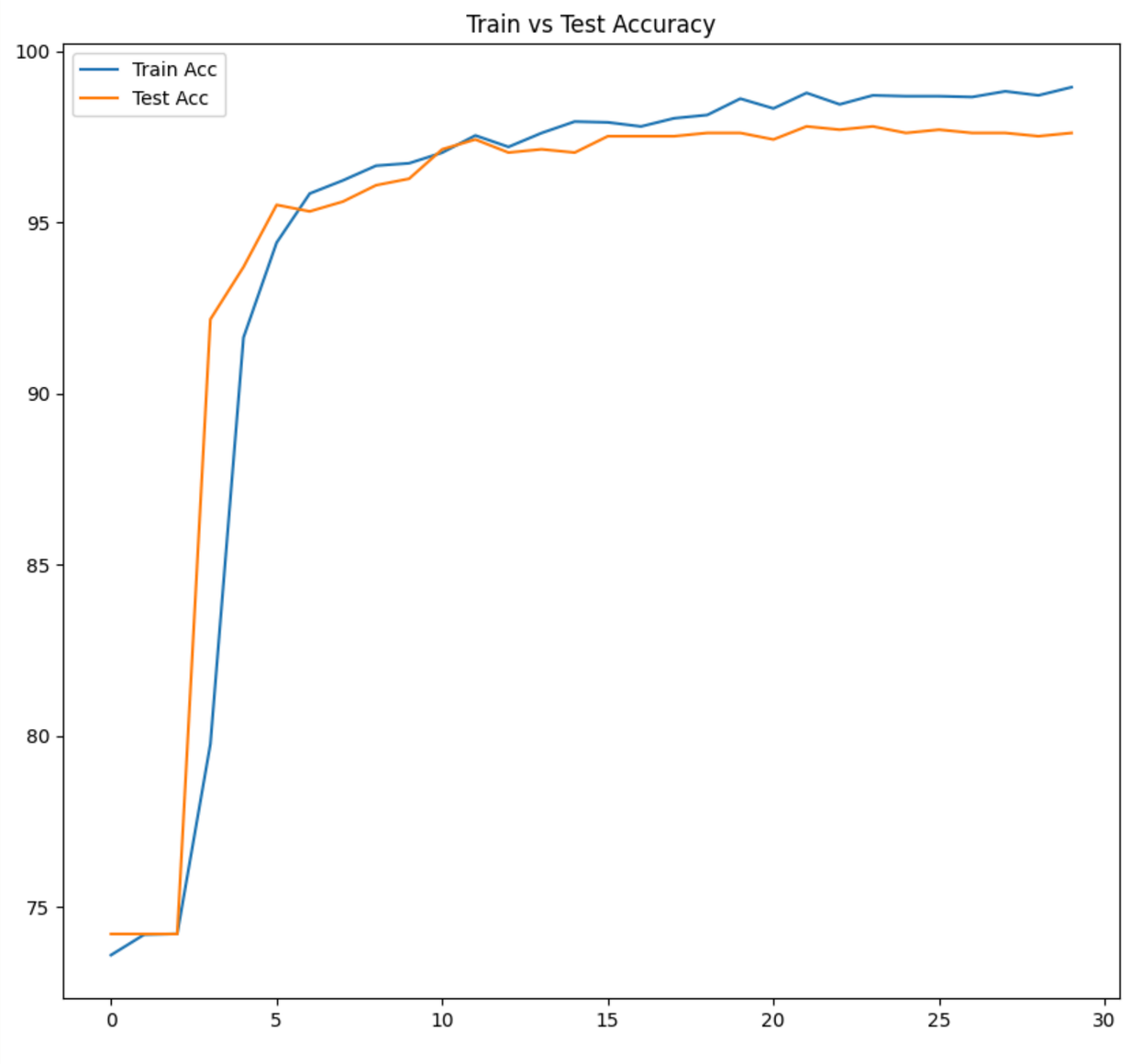
- **Robustness Test:** the test set has been modified by applying **gaussian blur**, **salt & pepper noise** and **speckle noise** to see how the prections change on noisy data

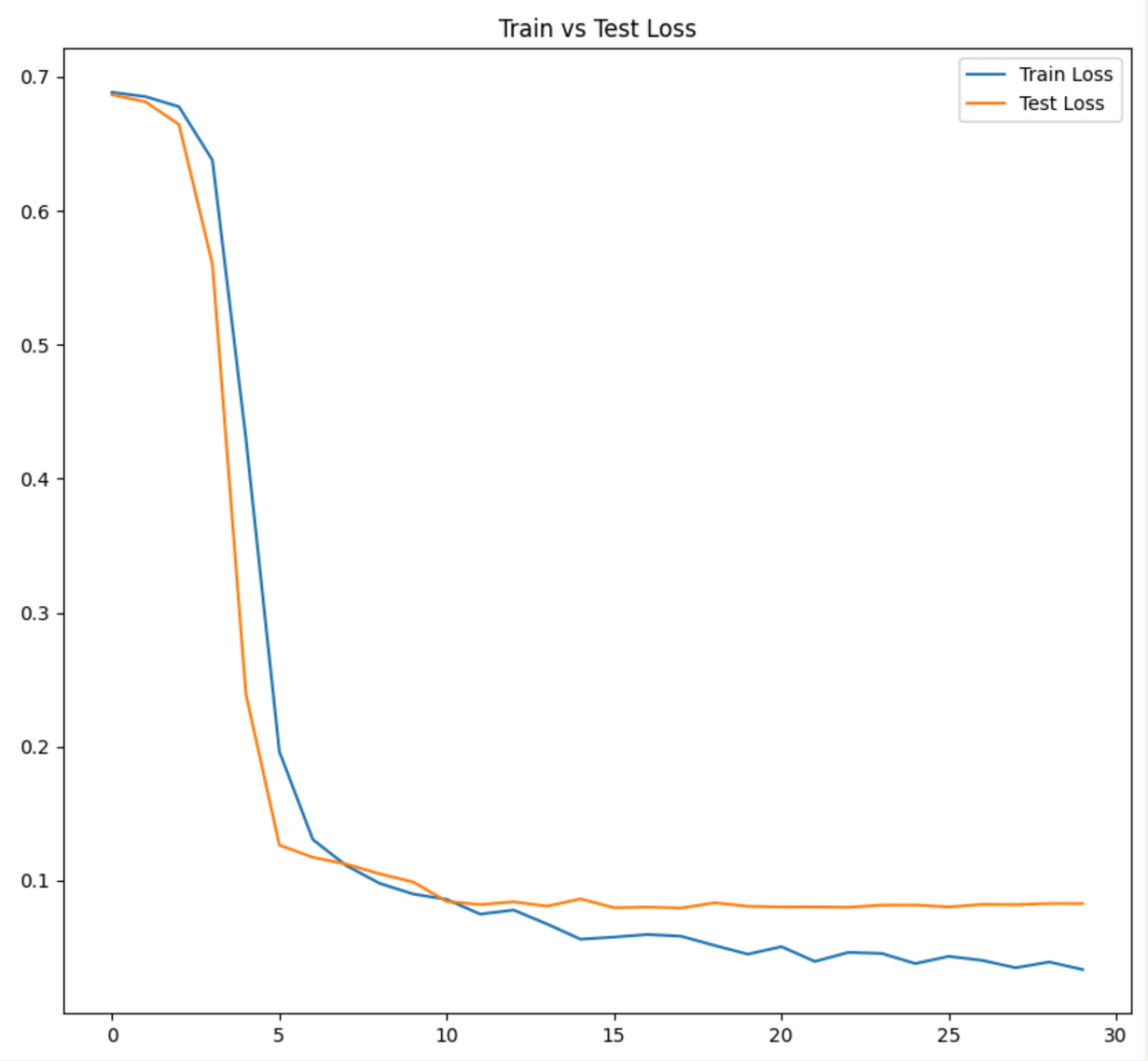
EXPERIMENTAL RESULTS

- **CROSS-VALIDATION:** the model with the lowest validation loss is the NN with the BiGRU trained with lr=1e-5 and beta=0.95.

However, the model without GRU has almost the same validation loss as the best model found (0.1130 vs 0.1221), meaning that the BiGRU is not that meaningful.

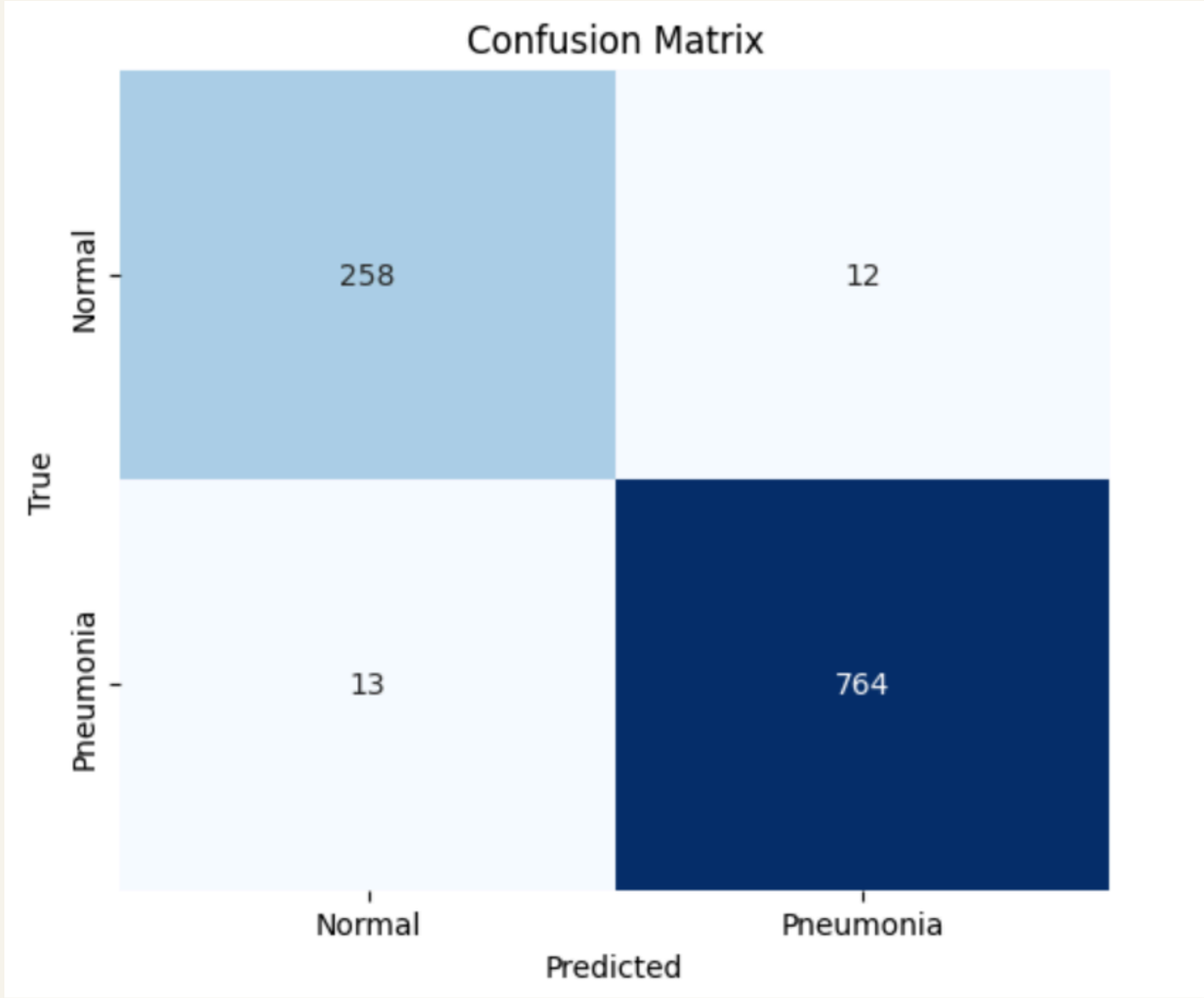
EXPERIMENTAL RESULTS





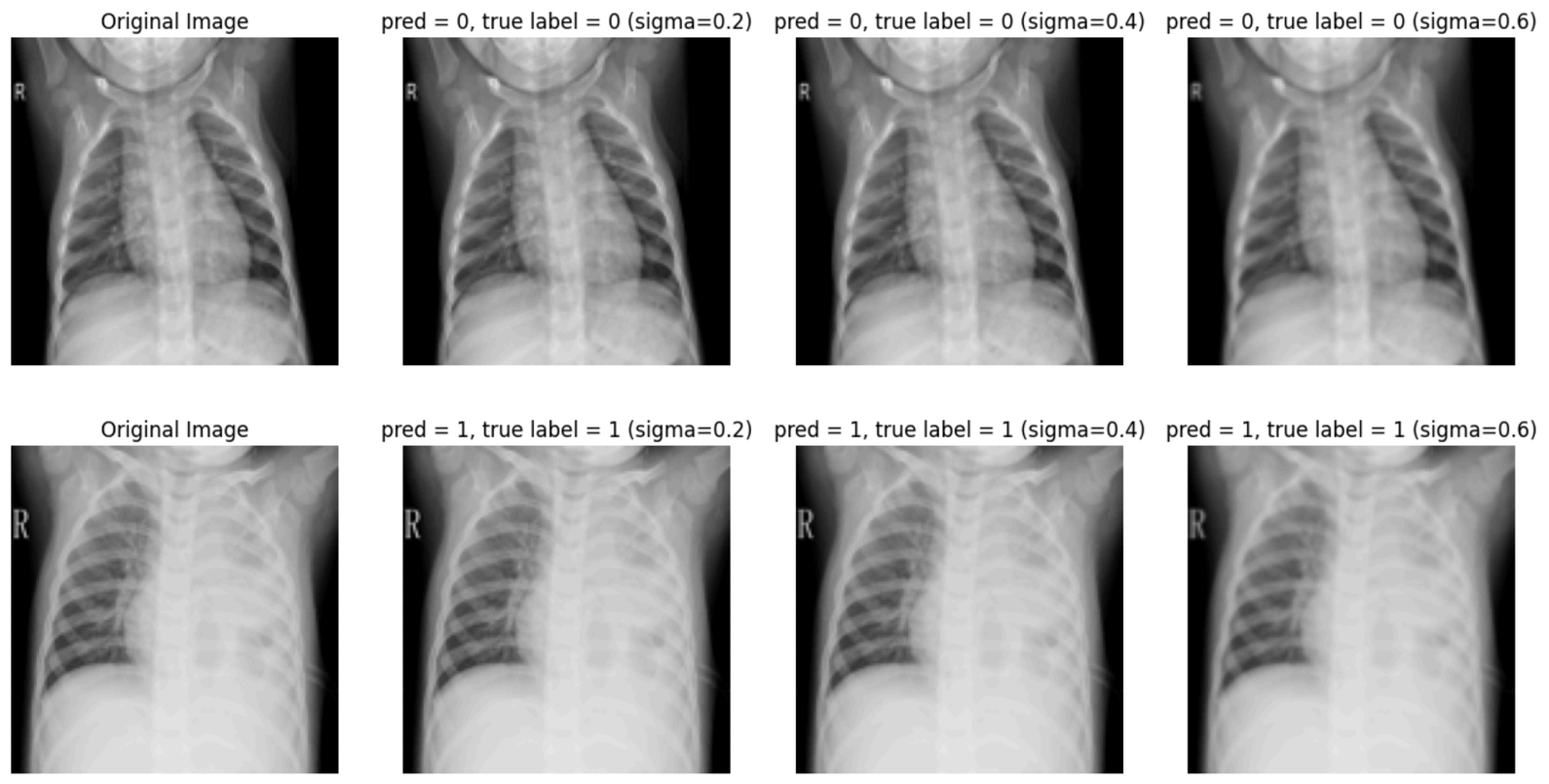
EXPERIMENTAL RESULTS

Metric	Value
Training Loss	0.0335
Test Loss	0.0828
Test Accuracy	97.61%
Precision	0.9845
Recall	0.9833
F1-Score	0.9839



EXPERIMENTAL RESULTS

GAUSSIAN BLUR



Discrepancies:

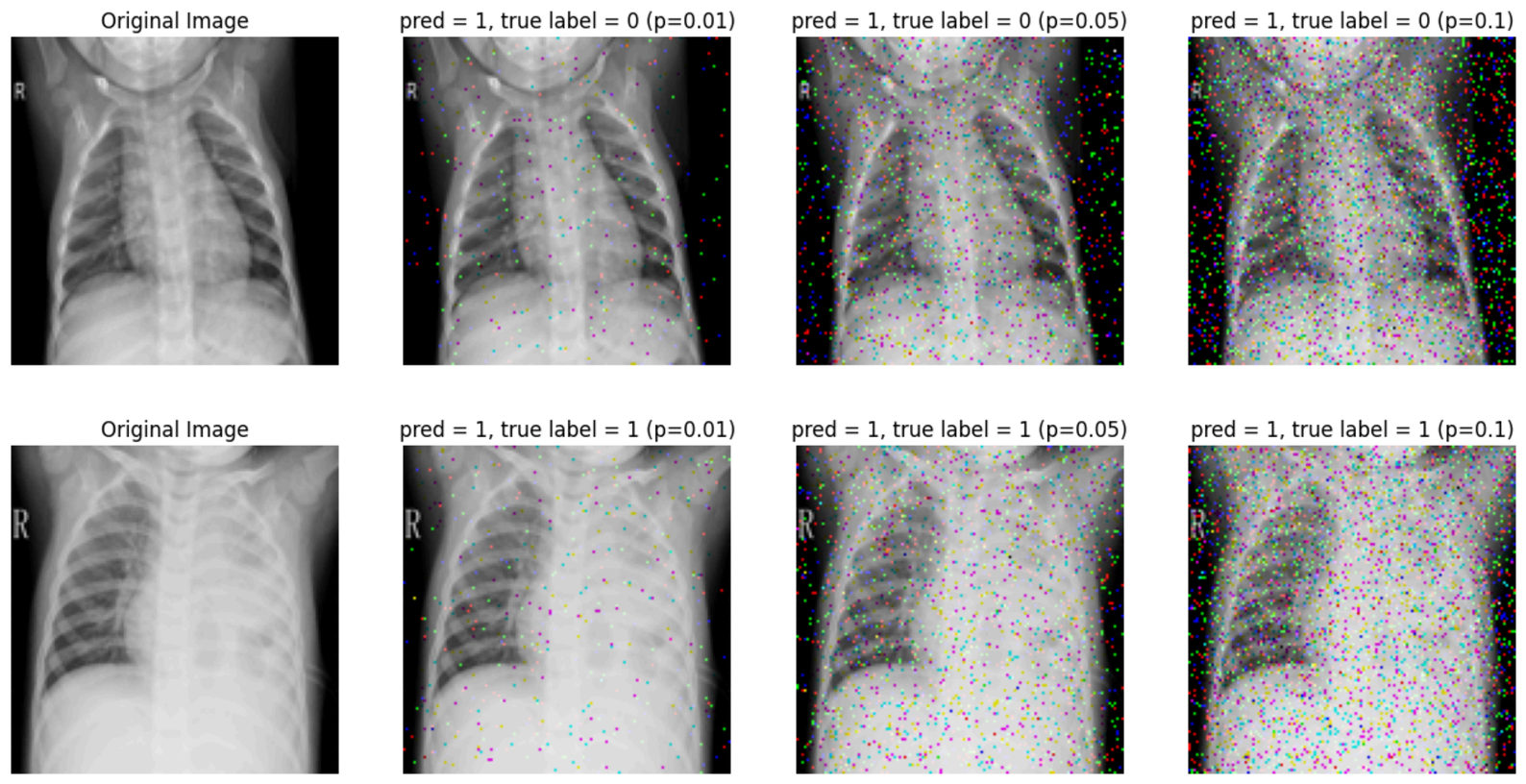
sigma = 0.2 -> 0.00%

sigma = 0.4 -> 0.00%

sigma = 0.6 -> 1.24%

EXPERIMENTAL RESULTS

SALT & PEPPER NOISE



Discrepancies:

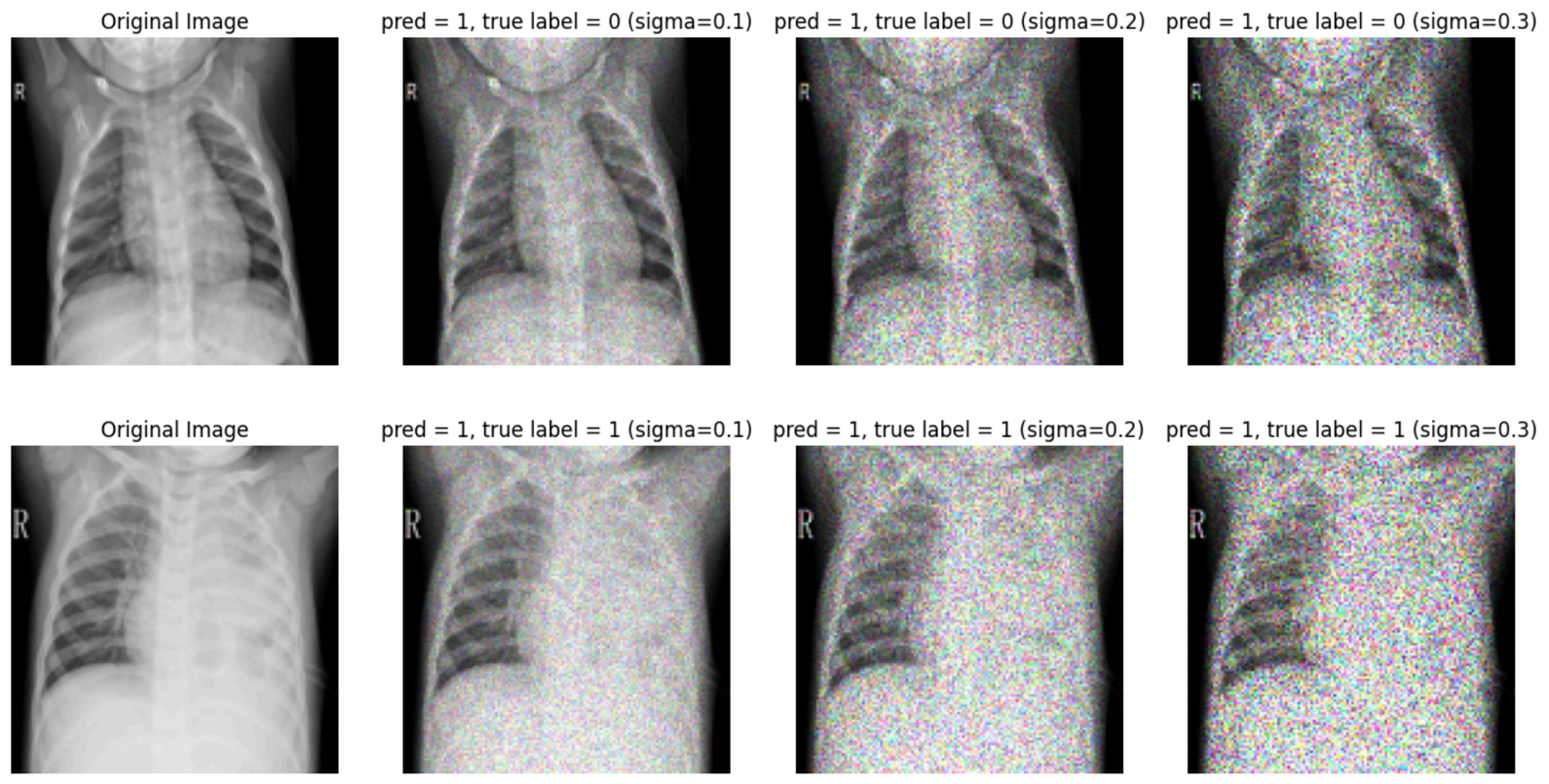
p = 0.01 -> 17.57%

p = 0.05 -> 25.79%

p = 0.1 -> 25.88%

EXPERIMENTAL RESULTS

SPECKLE NOISE



Discrepancies:

sigma = 0.1 -> 11.27%

sigma = 0.2 -> 25.79%

sigma = 0.3 -> 27.79%

CONCLUSIONS

- **High diagnostic accuracy**
- **Marginal performance gap** between full architecture and model without BiGRU
- **Robust** with respect to blurring but quite **sensitive** to speckle and salt & pepper noise